

FORENSIC EXAMINATION REPORT

Case Number:	001
Case Name:	Digital Forensics Lab
Examiner:	Paulo Vaz — Digital Forensics Analyst
Institution:	Aissa Tecnologia da Informacao
Examination Date:	June 7-13, 2026
Report Date:	June 13, 2026

Digital Forensics Mentorship Program — Aissa Tecnologia da Informacao
Produced in a controlled laboratory environment for educational and portfolio purposes.

1. Object of Examination

The object of this examination is a 200MB raw disk image (evidence.dd) and its EnCase-format counterpart (evidence_e01.E01), created in a controlled laboratory environment to simulate a suspect device involved in a financial fraud scenario.

The examination was commissioned to answer the following forensic questions:

- 1. Are there deleted files on the device that may constitute evidence of fraud?
- 2. Is it possible to recover the content of deleted files?
- 3. Do file system timestamps confirm the incident's time window?
- 4. Are there signs of deliberate evidence destruction?

2. Methodology

This examination was conducted in accordance with internationally recognized digital forensics standards and best practices: ISO/IEC 27037:2012, ISO/IEC 27041/27042/27043, RFC 3227, and NIST SP 800-86.

Tools Used

Tool	Version	Purpose
dd	GNU coreutils	Raw disk image creation
ewfacquire / ewfverify	libewf 20140816	E01 acquisition and verification
md5sum / sha256sum	GNU coreutils	Integrity hashing
The Sleuth Kit (mmls, fls, istat, icat, blkls, mactime)	4.14.0	File system analysis
foremost	1.5.7	Signature-based file carving
exiftool	13.55	Metadata and EXIF extraction
strings + grep	GNU binutils	Raw string search
Autopsy	4.23.1	Integrated graphical analysis

Chain of Custody

Cryptographic hashes were calculated immediately after image creation and verified prior to analysis. Any alteration to the image would produce a different hash value, invalidating the evidence.

Algorithm	Hash Value
MD5	5b98c6bdd1c195a11c8e2d6958dab25c
SHA256	d8d0d4b942f4dfd433189ae64aa4058599b5587a3018b210eb6c5bcb11d6edee

Both hashes verified before and after analysis: md5sum -c and sha256sum -c returned OK.

3. Evidence Examined

Item	Description	Format	Size	Integrity
evidence.dd	Raw forensic disk image	RAW (.dd)	200 MB	Verified
evidence_e01.E01	EnCase format image	E01	201 MB	Verified

4. Technical Analysis

4.1 Partition Analysis

Partition layout identified using mmls:

```
002: 000:000 0000002048 0000409599 Linux (0x83)
```

A single Linux (ext4) partition was identified, starting at sector 2048. All subsequent analysis used this offset.

4.2 File System Analysis

Active files identified via fls:

Inode	Name	Status	Size
11	lost+found	Allocated	12288 bytes
15	foto_reuniao.png	Allocated	442 bytes
16	log_sistema.log	Allocated	233 bytes

Deleted files identified via fls -r:

Inode	Original Name	Status
13	transfer_receipt.txt (OrphanFile-13)	Deleted
14	credentials.txt (OrphanFile-14)	Deleted

4.3 Inode Metadata Analysis

Metadata inspection via istat for deleted inodes:

Field	Inode 13	Inode 14
Created	2026-06-07 12:26:08	2026-06-07 12:27:03
Deleted	2026-06-07 12:33:09	2026-06-07 12:33:09
Size	0 (blocks zeroed)	0 (blocks zeroed)

Although file content was not recoverable via icat (blocks zeroed by the VirtualBox shared folder filesystem), the timestamps constitute valid forensic evidence that both files existed and were deliberately deleted at 12:33:09.

4.4 Raw String Search

Direct binary search via strings + grep recovered the following data from the raw image:

```
Transferencia PIX R$ 5000 CPF 123.456.789-00  
2026-06-07 12:24:03 TRANSFERENCIA PIX valor=5000
```

Search of 181MB of unallocated space (extracted via blkls) recovered:

```
senha_acesso: admin@2024
```

These findings confirm that file deletion did not erase the underlying data — content remained physically present on the disk and was recoverable through raw binary analysis.

4.5 File Carving

Foremost performed signature-based carving across the entire 200MB image:

```
0: 00020486.png 442 B offset 10488832 (640 x 480)  
1 FILES EXTRACTED
```

One PNG file was recovered at byte offset 10,488,832. The file is intact and valid (640x480 pixels, 442 bytes), corresponding to the image file present on the device.

4.6 Metadata Analysis

ExifTool analysis of the recovered PNG file:

Field	Value	Significance
Modify Date	2026-06-07 16:29:38 UTC	Consistent with file creation in Phase 2
File Modification Date	2026-06-09 17:41:12	Written by Foremost during carving
GPS Coordinates	Not present	File was system-generated
Make / Model	Not present	File was system-generated

No temporal anomalies detected. Timestamps are internally consistent.

4.7 Forensic Timeline

Chronological reconstruction via fls -m and mactime:

Time (EDT)	Event	Significance
12:12:15	lost+found created	Filesystem initialized
12:26:08	OrphanFile-13 created	transfer_receipt.txt written to disk
12:27:03	OrphanFile-14 created	credentials.txt written to disk
12:29:38	foto_reuniao.png — full MACB	Image file created
12:32:00	log_sistema.log — full MACB	System log generated

12:33:09	OrphanFile-13 and 14 deleted (m.c.)	Both files deleted simultaneously
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The timeline establishes a clear sequence: two sensitive files were created between 12:26 and 12:27, and deleted 7 minutes later at 12:33 — consistent with deliberate evidence destruction.

4.8 Autopsy Consolidated Analysis

All findings were confirmed through independent analysis in Autopsy 4.23.1. Deleted Files view identified OrphanFile-13 and OrphanFile-14 with matching timestamps. File Metadata panel confirmed inode 15 with MD5 and SHA256 hashes. Visual timeline corroborated the chronological sequence. Full HTML report preserved at laudo/autopsy_report/report.html.

5. Conclusions

1. Are there deleted files that may constitute evidence of fraud?

Yes. Two files — `transfer_receipt.txt` (inode 13) and `credentials.txt` (inode 14) — were identified as deleted. Their existence and deletion are confirmed by inode metadata, forensic timeline, and Autopsy analysis.

2. Is it possible to recover the content of deleted files?

Partially. Inode-based recovery via `icat` was unsuccessful due to block zeroing by the VirtualBox shared folder filesystem. However, file content was recovered through raw string search and unallocated space analysis, confirming PIX transfer data, a CPF number, and credential information.

3. Do file system timestamps confirm the incident's time window?

Yes. MACB timestamps establish that both sensitive files were created between 12:26 and 12:27 on June 7, 2026, and deleted at 12:33 on the same date.

4. Are there signs of deliberate evidence destruction?

Yes. The simultaneous deletion of both files at 12:33:09 — 7 minutes after creation — is consistent with deliberate evidence destruction. This behavioral pattern is commonly observed in fraud investigations.

6. Limitations

- Inode-based content recovery (`icat`) was unsuccessful due to block zeroing by the VirtualBox shared folder filesystem (`vboxsf`). This is an environment-specific behavior and does not reflect the methodology's capability on physical devices or native Linux filesystems.
- The evidence image was synthetically created for laboratory purposes. Findings are forensically valid within the scope of this controlled environment.
- No GPS or device metadata was present in the recovered image file, as it was system-generated rather than captured by a camera or smartphone.

7. Recommendations

- Preserve all evidence files with verified hash values.
- In cases involving physical devices, use a hardware write blocker during acquisition.
- For credential data recovered (`admin@2024`), initiate immediate access revocation procedures if applicable.
- Cross-reference PIX transfer data with financial institution records via legal request.

8. Examiner Declaration

I declare that this report was produced based solely on the evidence examined, using recognized digital forensics methodologies, and that the findings described herein are an accurate and objective representation of the technical analysis performed.

Examiner: Paulo Vaz — Digital Forensics Analyst

Institution: Aissa Tecnologia da Informacao

Date: June 13, 2026

This report was produced in a controlled laboratory environment for educational and portfolio purposes, as part of a Digital Forensics mentorship program. All procedures followed internationally recognized forensic standards.